SDN-Socket Programming – Jackfruit Mini Project

Goal:

The objective of this project is to design, implement, and evaluate a networked application using low-level TCP/UDP socket programming integrated with SDN. The project should demonstrate a clear understanding of TCP/UDP communication, concurrency, protocol design, programmable network control, and performance evaluation within a Mininet-based environment.

Mandatory Requirements:

- The application must be implemented using TCP or UDP sockets as specified in the project.
- The project must include appropriate client, server, worker, broker, or peer components based on the selected application.
- The application should support multiple communicating nodes/clients wherever applicable.
- The application must be deployed and tested in a Mininet-based network topology.
- The project should demonstrate interaction between the socket application and the SDN-controlled network.
- The complete system should run in a controlled Mininet + SDN controller + socket-programming environment.
- Students must provide the topology, source code, configuration, test cases, and instructions required to reproduce the results.

Final demo code should be shared in GitHub repo with complete documentation.

Evaluation Criteria

Deliverable	Component	Marks
D1 – 15	Problem Understanding & Architecture	**2**
	TCP/UDP Socket Implementation	**4**
	Mininet + SDN Implementation	**4**
	Socket–SDN Integration	**3**
	Initial Testing & Demo	**2**
	D1 Total	**15**
D2 – 25	Complete Application Functionality	**4**
	Dynamic SDN Functionality	**6**
	Challenging Scenario / Robustness	**4**
	Performance Evaluation & Comparison	**5**
	Documentation, Final Demo & Viva	**6**
	D2 Total	**25**
TOTAL		**40**

Detailed Rubric

Component	Marks	Evaluation Criteria
Problem Understanding & Architecture	**2**	Clear understanding of the assigned problem; well-defined objectives, system architecture, components, communication flow, and expected network behavior.
TCP/UDP Socket Implementation	**4**	Correct implementation of the specified TCP/UDP communication using low-level sockets; appropriate client/server communication, connection or datagram handling, concurrency, error handling, and application-level message exchange.
Mininet + SDN Implementation	**4**	Correct Mininet topology; functional SDN controller; appropriate flow rules; proper host/switch connectivity; meaningful implementation of the SDN functionality required by the project.
Socket–SDN Integration	**3**	Effective interaction between the socket application and SDN network; application/network events influence SDN behavior where required; end-to-end integration is demonstrated.
Initial Testing & Demo	**2**	Basic end-to-end system works; appropriate test cases are demonstrated; communication between application components and the SDN-controlled network is verified.
Complete Application Functionality	**4**	All major project requirements are implemented; application supports expected users/clients/flows; functionality works correctly under normal operating conditions.
Dynamic SDN Functionality	**6**	SDN controller dynamically responds to network/application conditions through mechanisms such as routing, filtering, prioritization, load balancing, rerouting, monitoring, blocking, or recovery.
Challenging Scenario / Robustness	**4**	At least one meaningful challenging scenario is demonstrated, such as congestion, packet loss, link failure, server overload, unauthorized access, or service failure; system responds appropriately.
Performance Evaluation & Comparison	**5**	Appropriate metrics are measured; experiments are conducted under multiple conditions; results are presented clearly; proposed approach is compared with a suitable baseline/static approach.
Documentation, Final Demo & Viva	**6**	Complete and well-structured documentation; architecture, implementation, testing and results are clearly presented; reproducible final demonstration; students demonstrate individual understanding through viva.